# Ethan Xu

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## Education

**University of California, Berkeley** — Berkeley, CA

*B.A. in Computer Science & B.S. in Applied Math, 4.0 GPA* — *Expected May 2029*

- **Regents' & Chancellor's Scholar**: Berkeley's highest merit scholarship (Faculty Nominee, top 1% of applicants)
- **Coursework**: Data Structures/Algorithms, Linear Algebra/Discrete Math, Multivariable Calculus, Structure and Interpretation of Computer Programs, Fundamentals of Data Science, Discrete Math & Probability Theory

## Technical Experience

**Coupang** — Sep 2026 – Present

*Software Engineering Intern* — *Mountain View, CA*

- Building a prototype **AI content pipeline** that turns seller photos and product data into complete e-commerce listings in **under 30** seconds each, by pairing computer vision on product images with LLM-generated copy
- Keeping output quality consistent across product categories by engineering **category-specific prompt templates** and an **image-quality scoring layer** that rates seller photos before generation, while coordinating team research

**Climformatics Inc.** — Sep 2026 – Present

*Machine Learning Researcher* — *Berkeley, CA*

- Built LightGBM time-series forecasting models using ERA5 data to predict hourly solar irradiance components (GHI, DNI, DHI) 24 hours ahead, **reducing forecast error by 23–30%** over weather data regression models
- Developing a downstream **PV power model** for Hanford, CA that converts irradiance forecasts into predicted solar panel output, trained on **20+ years** of NOAA observations processed into quality-controlled hourly data

**IPMD Inc., ASTAI Group** — May 2025 – Aug 2026

*Software Engineering Intern* — *San Mateo, CA*

- Built TensorFlow/Keras logistic regression models predicting price direction from OHLCV, Elliott Wave, candlestick, and sup/res signals, **outperforming the S&P 500 by 13%** cumulatively over a **10-year backtest**
- Designed an automated BeautifulSoup pipeline that scraped real-time Yahoo Finance articles scored sentiment on a transformer-based model, and fed daily sentiment scores and other technical signals as multimodal model inputs

**Data Analysis and Intelligent Systems Lab, University of Houston** — Jun 2025 – Sep 2025

*Lab Assistant, Dr. Christoph Eick* — *Houston, TX*

- Applied transfer learning to adapt a lunar **horizon-classification CNN** to deep-sea imagery, freezing early convolutional layers and fine-tuning the classifier head to detect front-facing horizon frames across **1,000+ images**
- Mapped and Geospatially aligned NOAA bathymetric data at **3 arc-second resolution** with deep-sea horizon images assigning per-image depth and geofencing by region to build labeled **Stable Diffusion training datasets**

## Publications & Projects

**SENTINEL: Sentiment Evolution and Narrative Tracking in Extended LLM Interactions**

- Designed a framework to quantify **intra-textual sentiment drift** in long-context LLM interactions and agentic workflows, finding drift is **domain and model dependent**, with negative texts neutralizing over time
- Analyzed 10,000+ outputs from frontier LLMs via OpenRouter and Hugging Face, introducing a **Drift Propagation Index** and token-level attribution showing sentiment drifts in oscillatory bursts, not linearly
- Co-first authored and Published at the **NeurIPS MTI-LLM 2025 Workshop** [NeurIPS] [OpenReview]

**Fintrax (Project)** | *Python, Pandas, Hugging Face Transformers, Beautiful Soup*

- Developed a **FinBERT model** classifying confidence into strong, weak, and neutral bins, scoring based on sentiment discrepancies between prepared remarks and Q&A sections in earnings-call transcripts from Motley Fool
- Built a scrape, parse, label, train, and score pipeline sized for **53.6k transcripts containing 14M sentences**, handling 3 page layouts and with a lexicon-removed check showing the model more than the word list.

## Technical Skills

**Languages** : Python, Java, C++, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS
**Libraries & Software**: pandas, NumPy, scikit-learn, Matplotlib, OpenCV, Beautiful Soup, React, Next.js, Node.js, Flask, FastAPI, TensorFlow, Keras, PyTorch, Git, Json, Microsoft Office, Google Collab, Fireworks, Open Router
**Databases & Cloud**: PostgreSQL, MongoDB, Redis, Supabase, Firebase, AWS